
Tertiary Treatment

Disinfection · Membranes · CETP · Sludge

Wastewater and Solid Waste Management

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Secondary treatment removed 90% of the BOD.

So why is the water still not fit to discharge?

What survives biological treatment

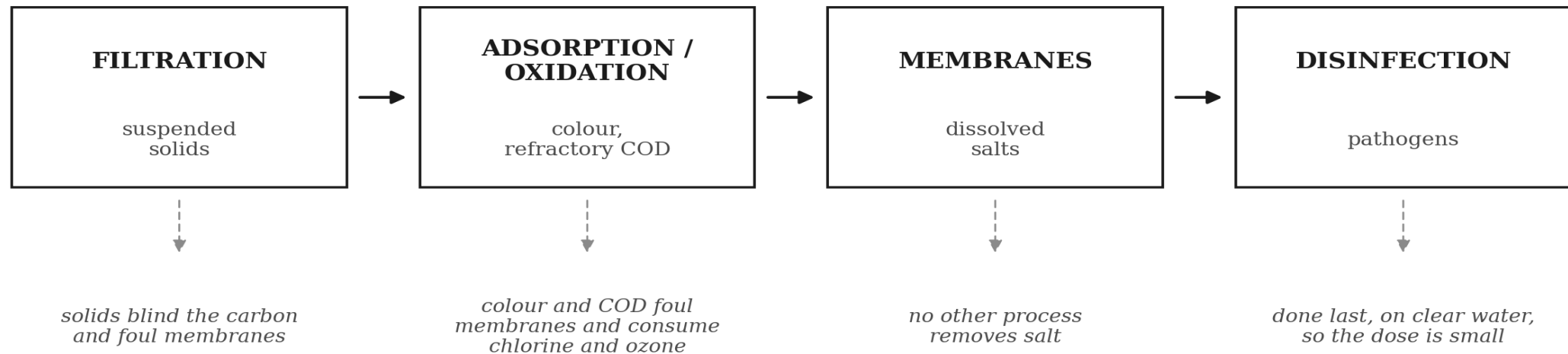
Four things pass through the secondary stage essentially untouched. Each needs a different answer, and between them they define the whole of tertiary treatment.

Residual suspended solids	20–100 mg/L of biological floc escaping the clarifier — and it carries BOD with it
Colour and refractory COD	dyes, syntans, vegetable tannins: microbes cannot digest them, so they pass through unchanged
Dissolved salts	biological treatment does not touch salt at all. A tannery effluent may carry 10–40,000 mg/L against a limit of 2,100
Pathogens	secondary removes 90–99%, which still leaves many thousands per millilitre

The third is the one that decides tannery design.

The order of the units

Figure 1. Order of tertiary units, and why each precedes the next



Each unit exists partly to protect the one that follows it.

What the law requires

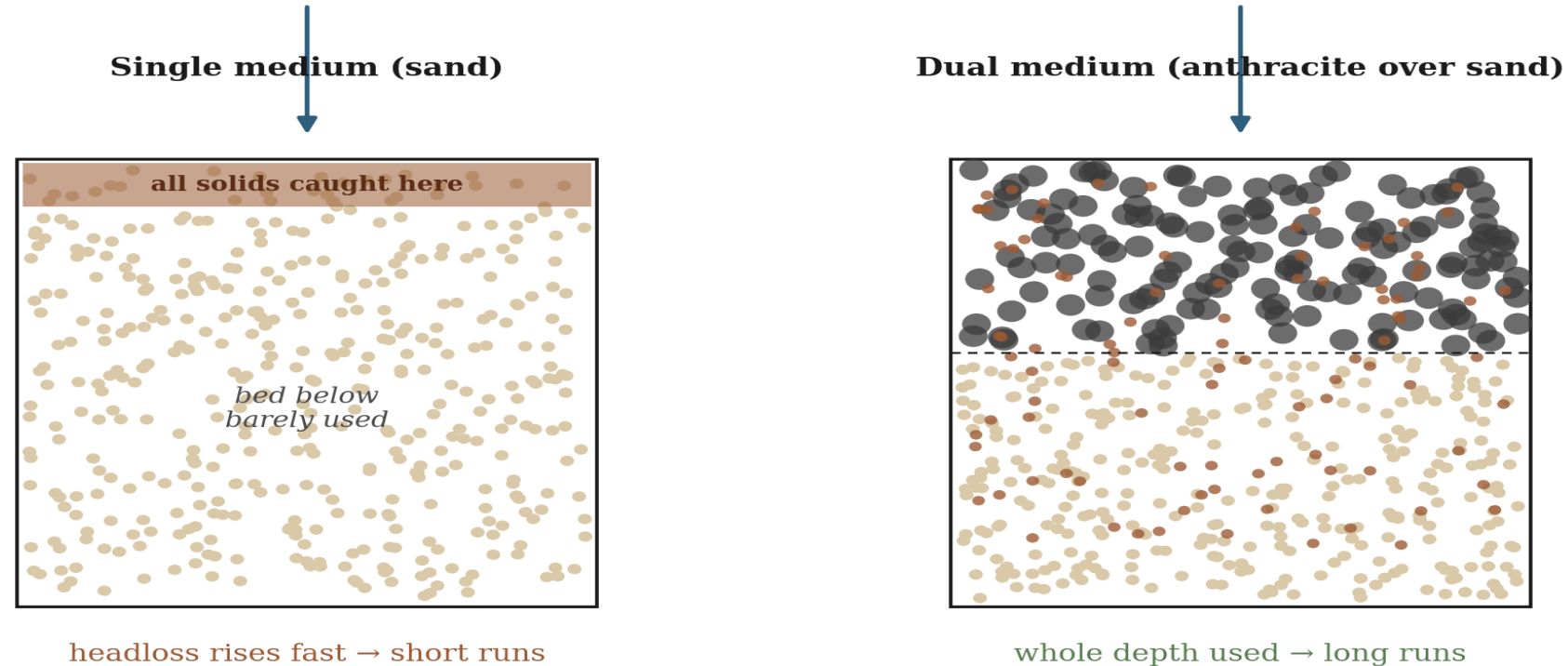
ECR 2023, discharge to inland surface water. The parameters that concern tertiary treatment:

Parameter	Limit	Usually met by
Suspended solids	150 mg/L	Filtration
BOD ₅	50 mg/L	Secondary; filtration assists
COD	200 mg/L	Activated carbon or oxidation
Colour	150 Hazen	Carbon, ozone, coagulation
Total dissolved solids	2,100 mg/L	Reverse osmosis — or salt reduction at source
pH	6 – 9	Neutralisation
Total chromium	2.0 mg/L	Recovered in primary treatment

After a good secondary stage a tannery usually complies on BOD, is marginal on COD and colour, and fails on TDS by a wide margin.

Filtration – the foundation of the train

Figure 7. Why dual media outlasts single media



Water meets COARSE media first, so solids penetrate instead of blinding the surface.

Backwashing re-sorts the bed: light anthracite settles back above the dense sand.

Disinfection

chlorination · ultraviolet · ozonolysis

One equation governs all three methods

Disinfection is not instantaneous. Organisms die progressively, and the rate depends on how much disinfectant is present and for how long.

Chick–Watson

$$C \times t = \text{constant}$$

concentration × contact time — the CT value

A low dose held for a long time and a high dose held briefly achieve the same kill. This is why every disinfection unit is really a contact tank, and why short-circuiting in that tank is so damaging: water that crosses quickly receives only a fraction of the design CT, and leaves untreated.

Log removal: 1 log = 90% · 2 log = 99% · 3 log = 99.9% · 4 log = 99.99%

Going from 99% to 99.99% sounds like one percentage point. It is a hundredfold reduction in survivors.

Six things that spoil disinfection

Suspended solids	shield organisms from the disinfectant; absorb UV. The worst offender of the six
Organic matter	consumes chlorine and ozone chemically — the dose is spent on the water, not the bacteria
Colour and turbidity	absorb and scatter UV light
Ammonia	converts chlorine to chloramines, which are perhaps a hundred times weaker
pH	decides whether chlorine exists as the strong form or the weak one
Temperature	warmer water kills faster

Every one of them is present in tannery effluent. This is why disinfection is placed LAST — after the water is clear.

Chlorination — the chemistry that matters



Both HOCl and OCl⁻ count as free available chlorine, and both appear on a laboratory report as the same number. But they are not equivalent at all: hypochlorous acid is 40 to 80 times the more effective, because it carries no charge and passes readily through the cell wall, while the negatively charged hypochlorite ion is repelled by the negatively charged cell surface.

Which of the two you have depends entirely on pH — and that is the single most useful fact in chlorination practice.

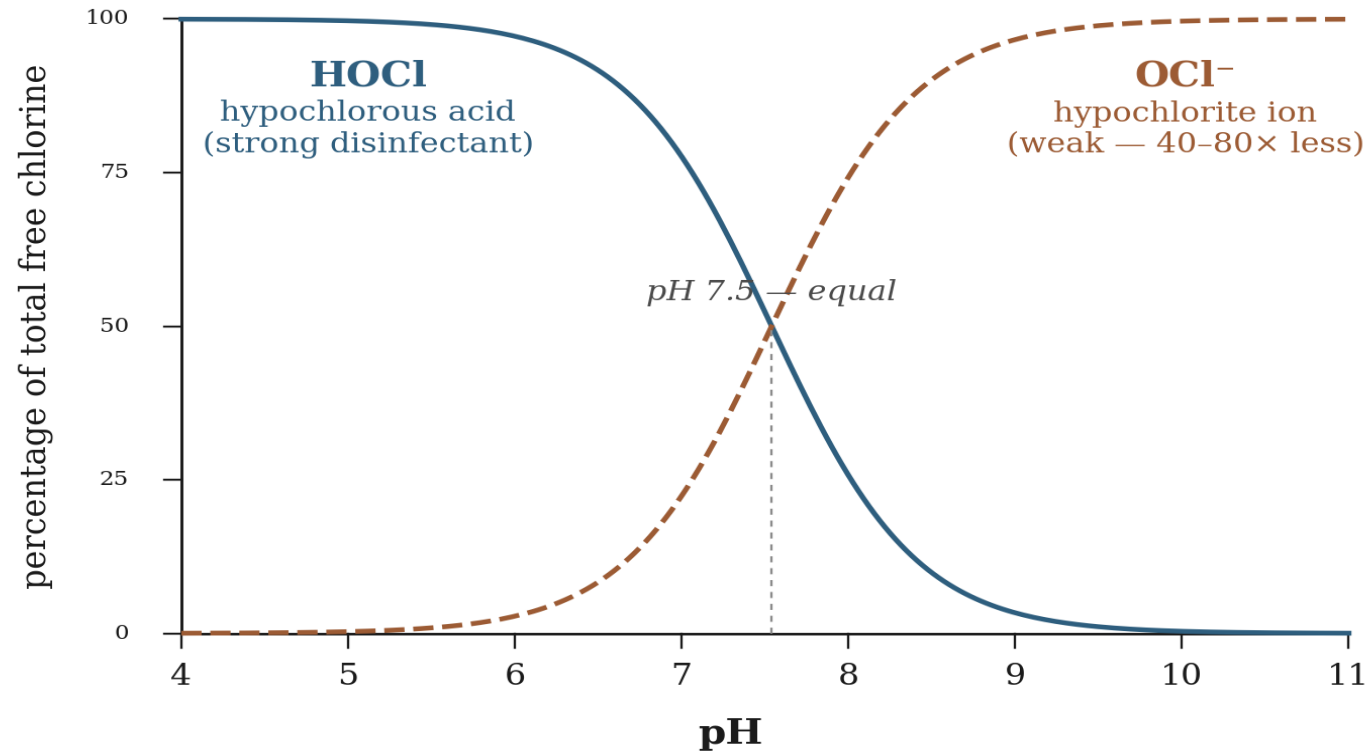
Below pH 6 — almost all HOCl

At pH 7.5 — half and half

Above pH 9 — almost all OCl⁻

Chlorine speciation with pH

Figure 2. Distribution of free chlorine with pH



Chlorination is markedly less effective in alkaline water.

Why chlorine is not the whole answer

- Trihalomethanes and other by-products form when chlorine meets residual organic matter — and a coloured tannery effluent is full of it
- The residual that protects the water downstream is itself toxic to fish, so a sensitive receiving water requires dechlorination afterwards
- Weak against protozoan cysts, whatever the dose
- Chlorine gas is acutely dangerous to store and handle
- In a high-ammonia effluent the breakpoint demand makes it expensive

Against all of that stands one merit no other method has:
chlorine leaves a residual, and goes on protecting the water after it has left the works.

Ultraviolet — disinfection without a chemical

Sunlight has always disinfected water. UV treatment simply concentrates the effect: lamps produce light at 254 nm, which is very close to the peak absorbance of DNA. A photon absorbed by DNA fuses adjacent thymine bases into a dimer. The strand can no longer be read or copied. The organism may remain metabolically alive, but it cannot reproduce — and an organism that cannot reproduce cannot infect anyone.

Nothing is added to the water. Nothing is left behind.

Doses

Wastewater disinfection
20–40 mJ/cm²

Water reuse
40–100 mJ/cm²

4-log virus
100–186 mJ/cm²

Protozoan cysts
10–20 mJ/cm²

*UV beats chlorine decisively
on cysts.*

The difficulty, for us

UV only works if the light reaches the organism.

Transmittance at 254 nm is the measure. Above 65% UV works normally; below 50% it becomes uneconomic or simply ineffective.

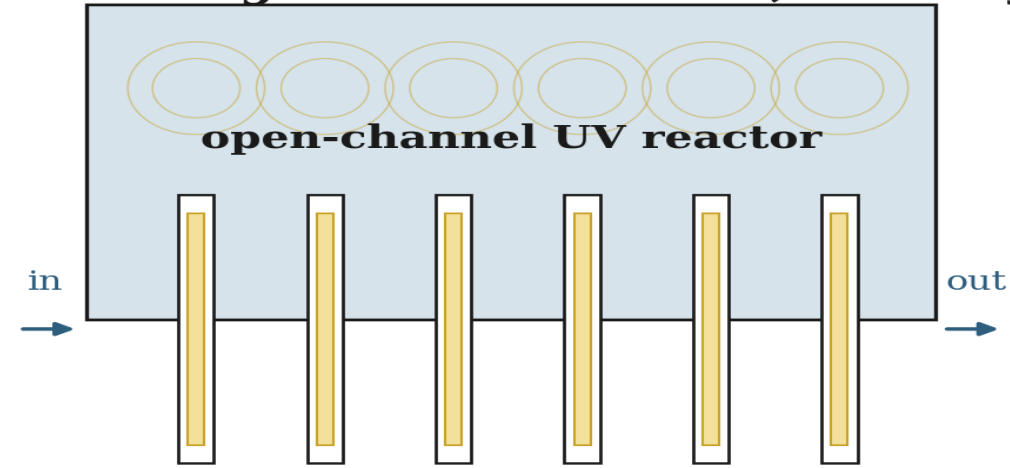
Tannery effluent is coloured, and colour absorbs ultraviolet strongly.

A dark effluent may transmit only 20–30%.

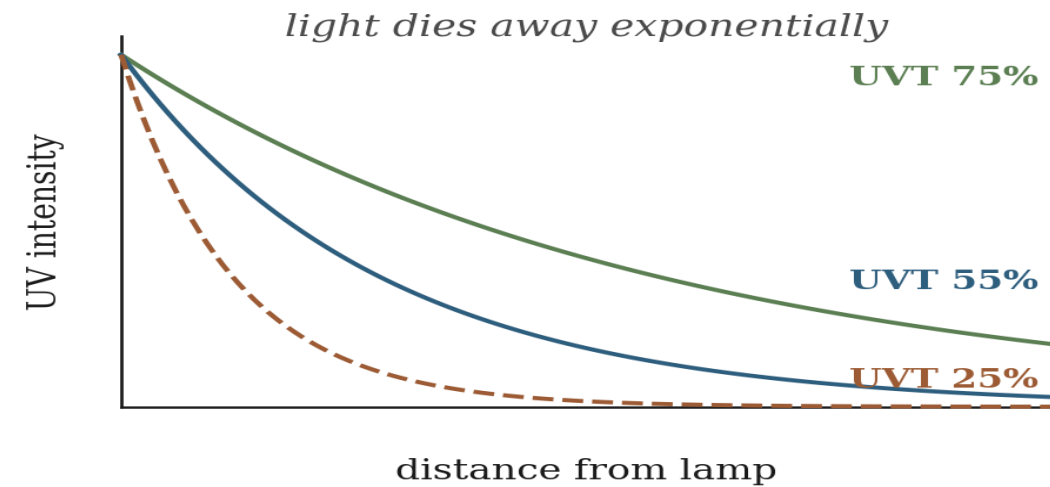
So UV is not ruled out for a tannery — but it can only come after the colour has been taken out by carbon or by ozone. Never before. And note the other catch: because UV damages DNA rather than destroying the cell, some organisms repair themselves afterwards in daylight. Photoreactivation is real, and the dose is set high enough to allow for it.

UV reactor and light penetration

Figure 8. UV reactor, and why transmittance decides everything



lamps in quartz sleeves · 254 nm · contact seconds



At 25% transmittance the light is gone within a few centimetres of the lamp.

Colour, not bacteria, is what defeats UV in a tannery effluent — so colour must be removed first.

Sizing a UV system

The dose is the product of intensity and exposure time — the direct analogue of the CT value.

$$\text{UV dose} = I \times t$$

$$\text{mW/cm}^2 \times \text{seconds} = \text{mJ/cm}^2$$

Required dose	wastewater disinfection	30 mJ/cm²
Lamp intensity	measured or from supplier	8 mW/cm²
Exposure needed	30 / 8	3.8 s
Flow	400 m ³ /d	4.63 L/s
Irradiated volume	4.63 × 3.8	17.6 L

But all of this assumes the light arrives. Halve the transmittance and the lamp count roughly doubles.

Ozone

Ozone is a far stronger oxidant than chlorine. It must be made on site, in a corona discharge, because it cannot be stored. It attacks organisms directly, and at higher pH it decomposes into hydroxyl radicals, which are stronger still.

But for a tannery the interest is not usually disinfection at all.

Ozonolysis means cleaving organic molecules. Ozone attacks exactly the carbon–carbon double bonds that make a dye molecule coloured — so it destroys colour rather than collecting it. That is a genuine alternative to activated carbon, and it is the reason ozone appears in tannery schemes at all.

	Activated carbon	Ozone
Mechanism	adsorption — transfers the pollutant	oxidation — destroys the molecule
Residue	spent carbon to regenerate or dump	none; reverts to oxygen
Capacity	finite — breakthrough occurs	continuous
Cost	lower capital, carbon replacement	high capital, high electricity

Ozone



Choosing between the three

Situation	Method
Discharge to a distant watercourse; residual wanted	Chlorination
Sensitive receiving water; no chemicals permitted	UV — if the effluent is clear
The effluent is coloured and colour must also go	Ozone
Protozoan cysts are the concern	UV
Ammonia is high	UV or ozone — chlorine would form chloramines
Budget is the overriding constraint	Chlorination
Water is to be reused after RO	UV, as a final barrier

Reverse Osmosis

the only answer to dissolved salt

The problem nothing else touches

Screening, settling, biological treatment, filtration, carbon, ozone — every process so far leaves the dissolved salts exactly as it found them.

10,000 – 40,000

mg/L TDS in a typical
tannery effluent

2,100

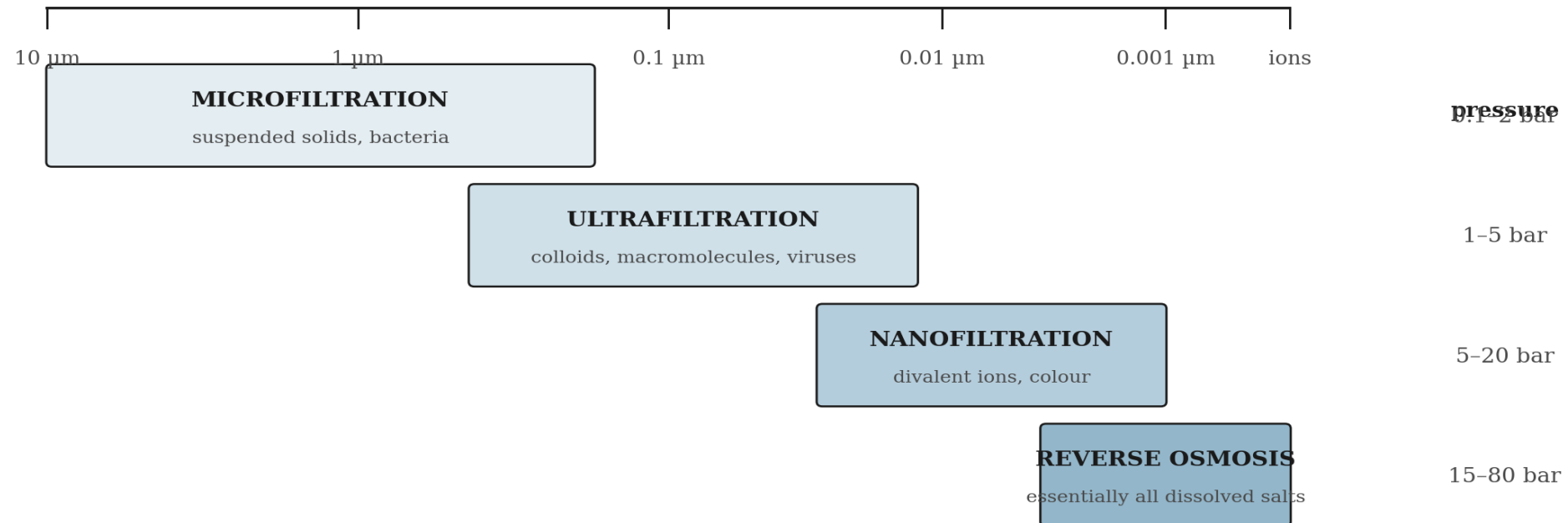
mg/L — the discharge limit

Salt is used lavishly in this industry — curing, pickling, chrome tanning — and all of it ends up in the effluent. There are exactly two answers. Either it is never used in the first place, which is cleaner production; or it is separated physically by a membrane, which is this section.

Where RO sits among the membranes

Figure 4. The membrane spectrum

decreasing particle size →



Only nanofiltration and reverse osmosis remove dissolved salt.

MF and UF exist to protect the RO from fouling.

Osmosis, reversed

Put fresh water and salt water either side of a membrane that passes water but not salt, and water moves of its own accord into the salt solution, diluting it. The pressure that would just stop that movement is the osmotic pressure.

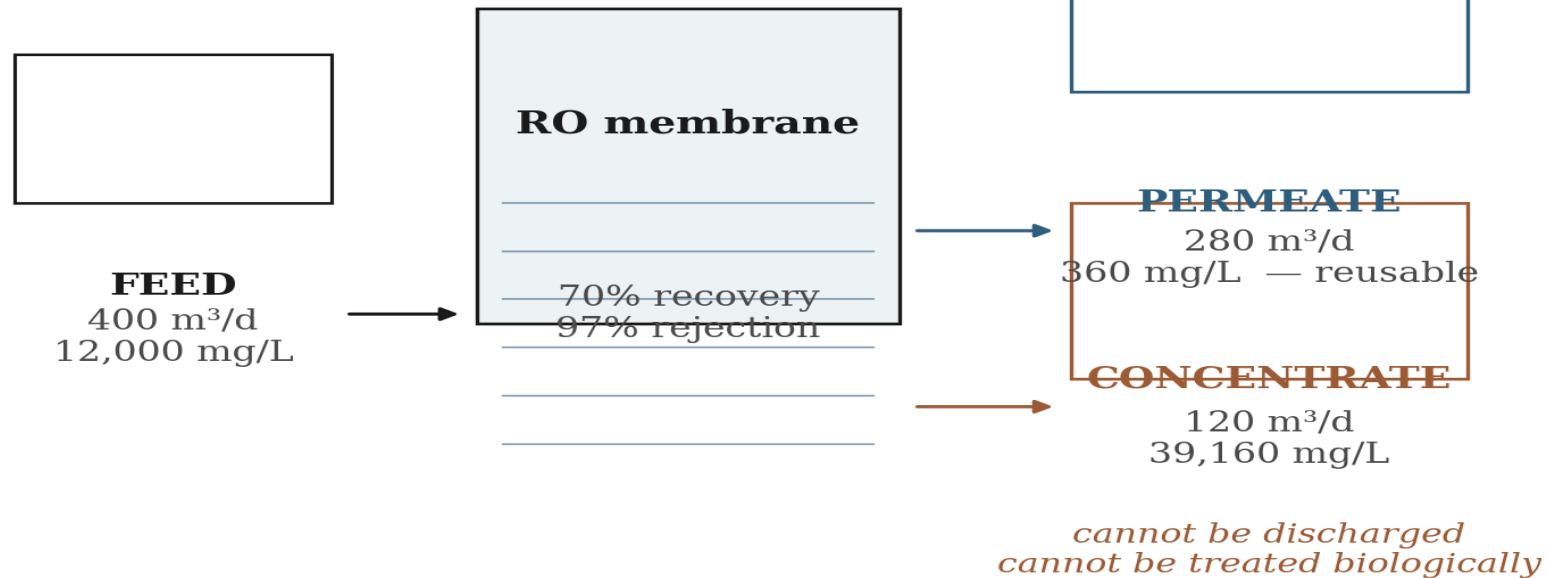
Apply to the salty side a pressure greater than the osmotic pressure and the flow reverses. Water is squeezed out of the salt solution, leaving the salt behind.

$$\pi \approx 0.077 \text{ bar per } 1,000 \text{ mg/L of TDS}$$

Note that this is not filtration in the ordinary sense. The membrane has no pores that salt is too big to enter; water dissolves into a dense polymer film and diffuses through it, while ions do not. That is why the pressures are so high — 15 to 80 bar — and why the energy cost is what it is.

Worked example – where the salt goes

Figure 5. Reverse osmosis – where the salt goes



RO does not destroy salt — it concentrates it.

The concentrate is the central difficulty of every reuse scheme.

And then what?

Reusable water — which was the object

Brine, which cannot be discharged and cannot be treated biologically.

Four options exist, and none is comfortable.

- Discharge to sea or deep well — where geography and law permit, which in Bangladesh they rarely do
- Evaporation ponds — simple, but large area and secure lining needed
- Mechanical evaporation and crystallisation to a solid salt — the zero-liquid-discharge route
- Use less salt in the first place, so that less arrives to be concentrated

Every kilogram of salt that enters the tannery must eventually leave it — in the effluent, in the concentrate, or as solid salt from a crystalliser. The cheapest place to deal with it is the curing yard.

The CETP

a shared plant for a cluster of tanneries

Why a common plant at all

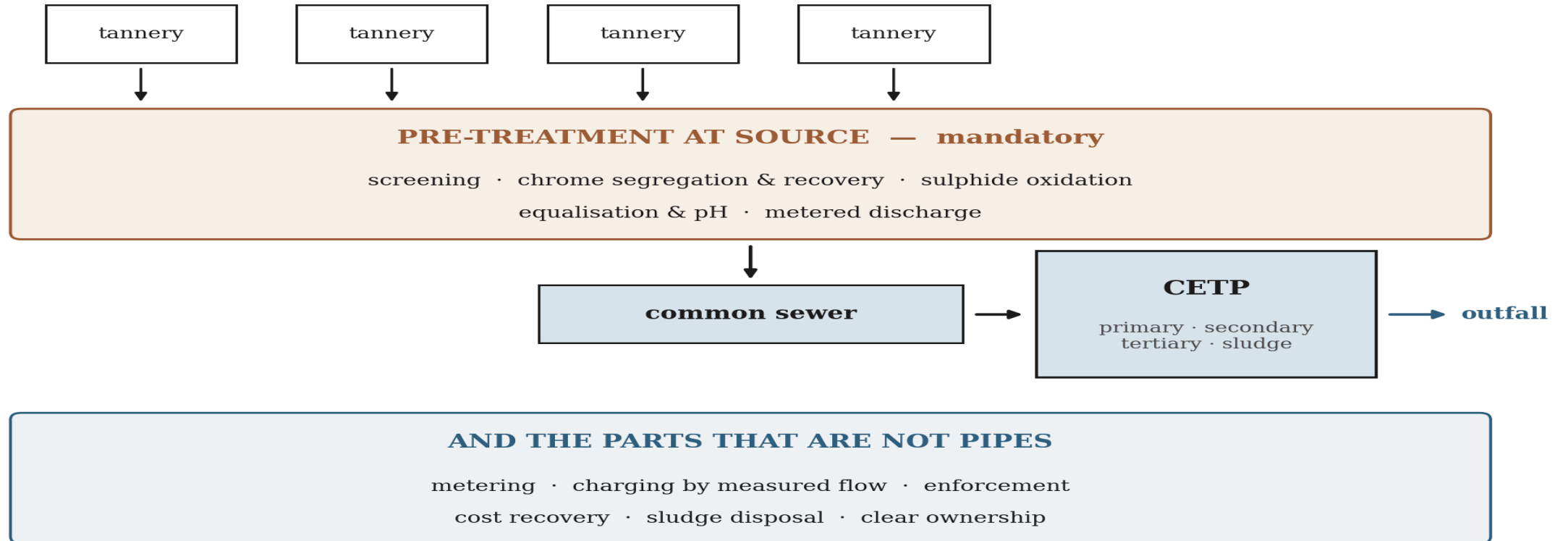
A tannery processing a few hundred hides a day cannot justify a full treatment works — not the land, not the capital, not a trained chemist. Yet the effluent of fifty such tanneries is exactly as damaging as that of one large one. The problem is structural, not technical.

One plant of 5,000 m³/d costs far less than fifty of 100 m³/d, and can afford what none of them could:

- proper equalisation, because fifty uncoordinated discharges average each other out
- a laboratory, trained operators and continuous monitoring
- a sludge-handling facility built to the standard required
- tertiary treatment, RO and chrome recovery at a scale that makes them economic
- a single outfall the regulator can meter and sample

What a CETP actually consists of

Figure 10. A CETP is a system, not a plant
member tanneries



Why they so often disappoint

Hydraulic overload	more tanneries connect than the design allowed for
No pre-treatment at source	chrome and sulphide arrive already mixed, when neither can then be dealt with
Highly variable influent	fifty different processes; no single design suits them all
Bypassing and night discharge	charges based on production rather than metered flow remove the incentive
Weak cost recovery	members do not pay, maintenance is deferred, equipment fails
Sludge neglected	chrome-bearing sludge is hazardous, and its disposal is expensive
Diffused responsibility	when everyone shares a plant, nobody owns its performance

A CETP can only be as good as the pre-treatment of the factories feeding it.

Savar — the numbers

The relocation from Hazaribagh to the Savar Tannery Industrial Estate, with a CETP at its centre, was meant to end the untreated discharge that had ruined the Buriganga.

	Designed for	Actually
CETP hydraulic capacity	25,000 m ³ /d	assessed at 14,000 – 18,000 m ³ /d
Effluent generated	—	about 40,000 m ³ /d; 42,000 – 45,000 at Eid
Chrome recovery (CCRU)	1,050 m ³ /d	about 5,000 m ³ /d of chrome liquor arrives
Pre-treatment at source	mandatory for every tannery	reported as not being followed
Tertiary / RO for salt	—	not provided
Central sedimentation	—	not provided

Roughly Tk 1,015 crore over eighteen years; seventeen project directors, twelve of them part-time.

Savar — what it teaches

Read that table against the previous figure. Almost none of the failures are failures of the treatment plant as a piece of engineering.

- Capacity was outrun by the load connected to it — a planning and enforcement failure
- Chrome arrives mixed because pre-treatment at source is not enforced — so the recovery units, already undersized fivefold, cannot work at all
- The sludge is therefore chrome-bearing, and becomes hazardous waste rather than recovered raw material
- Salt was never addressed, because no RO was built — a decision taken at design stage
- Eighteen years and seventeen project directors is a governance problem, not a process one

The concept is not unsound. A CETP is a system, and the system was never completed.

Figures from press and published assessments, 2021–2026; they move, so check before quoting.

In-plant management

housekeeping · segregation · water · chemicals · reuse

Housekeeping

The cheapest interventions in this course. None needs capital equipment; all need supervision. Together they are routinely worth 10–25% of water and chemical load.

Measure the float	drums run to a marked ratio, not to a hose left open
Stop the running hose	trigger nozzles and timed valves — an open 25 mm hose wastes ~25 m ³ a shift
Contain the spills	drip trays under drums, kerbed stores, bunded pits
Fix the leaks	worn drum seals, dripping valves and overflowing pits are unmetered load
Meter and record	a meter per section and a daily logbook — nothing unmeasured is reduced
Train and supervise	same drum, same recipe, two operators, twice the water

Spilt chemical is bought twice — once from the supplier, once from the CETP.

Segregation of waste streams

Four streams must not meet before treatment. Mixing destroys the option to recover any of them — and the mixture is harder to treat than any of its parts.

Stream	Where it comes from	If kept separate	If mixed
Chrome liquor	spent tanning and wet-back floats	precipitated and returned to the tanyard	chrome enters all sludge — hazardous waste
Sulphide liquor	lime—unhairing floats	hair screened, then catalytic oxidation	meets acid → H ₂ S; kills the biology
Salt-laden liquor	curing yard drainage, spent pickle	held for evaporation or reuse	10–40,000 mg/L TDS through the whole plant
Clean water	cooling, blowdown, roof runoff	reused directly or discharged	dilutes the effluent and inflates the plant

Segregation costs pipework and discipline. Mixing costs a hazardous classification on every tonne of sludge.

Reduction of water use

A conventional tannery uses 30–40 m³ per tonne of raw hide; good practice reaches 12–18. The difference is not one technology but a series of decisions about floats and washes.

Measure	How it works	Typical saving
Short floats	tanning and post-tanning at 40–80% float, not 150–200%	15–30% of process water
Batch washing	fill, run, drain — instead of a hose running as the drum turns	up to 50% of wash water
Fewer, better washes	two efficient washes beat five careless ones; check by drain conductivity	10–20%
Reuse of clean streams	cooling water, final rinses and RO permeate returned upstream	10–25%
Automatic drum controls	timed, metered filling — volume no longer operator-dependent	5–15%

Every m³ not used is a m³ not pumped, not heated, not treated and not discharged. The saving counts four times.

Worked example — water balance and savings

6 t of raw hide per day at 33 m³/t, discharging to the estate CETP at Tk 45 per m³. Short floats, batch washing and cooling-water reuse are introduced.

Present water use	$6 \text{ t/d} \times 33 \text{ m}^3/\text{t}$	198 m³/d
After the measures	$6 \text{ t/d} \times 20 \text{ m}^3/\text{t}$	120 m³/d
Water saved	$198 - 120$	78 m³/d (39%)
CETP charge avoided	$78 \times \text{Tk } 45$	Tk 3,510/d
Over 300 working days	$3,510 \times 300$	≈ Tk 1.05 million/yr
COD load, before	$198 \text{ m}^3 \times 2,500 \text{ mg/L}$	495 kg/d
COD load, after	$120 \text{ m}^3 \times 3,600 \text{ mg/L}$	432 kg/d

Watch the concentration. The load falls 13%, but the effluent strengthens from 2,500 to 3,600 mg/L — welcome to a biological stage, punishing to any unit that was sized on concentration.

Reduction of chemical use

Every kilogram that does not fix on the hide leaves in the effluent. Two levers only: offer less, and fix more of what is offered.

High-exhaustion chrome	controlled pH, temperature and basification: 60–70% exhaustion becomes 90–95%
Hair-save unhairing	hair screened rather than dissolved — cuts beamhouse COD by 30–50%
Low-salt curing	chilling, short-term preservation or biocides in place of 40% salt on hide
CO₂ deliming	replaces ammonium salts, removing ammonia nitrogen at source
Enzyme assistance	in soaking, unhairing and bating — lower chemical offer and shorter processes
Reduced sulphide	amine or enzyme-assisted systems, where hide quality permits

A chemical never bought is cheaper than one recovered, which is cheaper than one treated.

Recycling and reuse of chemicals

Four tannery liquors are genuinely recyclable at plant scale. Each depends absolutely on segregation — none of them survives being mixed.

Liquor	Treatment	Returned as	The condition
Spent chrome	precipitate with MgO or NaOH, settle, redissolve in H ₂ SO ₄	20–30% of the fresh chrome offer	tanyard stream kept separate
Spent lime–sulphide	screen the hair, settle, top up sulphide and lime	30–50% of the unhairing float	solids removed every cycle
Spent pickle	filter, then adjust salt and acid to strength	most of the pickle float	protein build-up limits the cycles
Spent vegetable liquor	counter-current pit system	progressive re-use of tannin	long-standing practice, not new

Chrome recovery pays twice — and sludge without chrome is not hazardous waste.

Recycling and reuse of water

Water need not be clean to be useful — only clean enough for the next job. The rule is to cascade: cleanest water to the most sensitive step, then pass it down.

Cooling and boiler water	never touched the hide — reuse directly, or send it to soaking
Final rinses	lightly loaded — return as the first wash of the next batch
Soak and wash liquors	screened and settled — back to the curing yard or the first soak
RO permeate	300–400 mg/L TDS — the only stream fit to return to a tanning float
Treated CETP effluent	floor washing, dust suppression, gardening — never into a float

The hierarchy is set by salt, not by BOD. Anything carrying TDS cannot go back into a float, however clear it looks.

Water footprint

ISO 14046. The footprint counts the water behind the product, not the water through the meter — and for leather it is dominated by a component the tannery does not control.

Blue	surface or ground water abstracted and not returned to the same catchment — the tannery's own 12–40 m ³ /t
Green	rainwater consumed growing the feed for the animal — dominant for any livestock product
Grey	the clean water needed to dilute the discharge back to standard — where treatment appears

≈ 17,000 litres per kg of bovine leather, cradle-to-gate.

The tannery's own abstraction is a few percent of that. Almost all of it is upstream livestock.

Blue water alone flatters the sector. Grey water is the component in-plant management actually moves.

Think about it

Tk 45 per m³ at the CETP. You use 33 m³ per tonne of hide.

Your managing director will fund exactly one of the following this year.

- Automatic float metering on every drum, and a written washing procedure
- A chrome recovery unit on the segregated tanyard stream
- Hair-save unhairing in the beamhouse
- A hose, nozzle and logbook discipline programme, with a supervisor accountable for it

Which pays back first, which reduces the CETP's problem most — and are they the same one?

Two of these cost almost nothing. That is not the same as being worth the least.

Sludge

what is left when the water is clean

Nothing has been destroyed

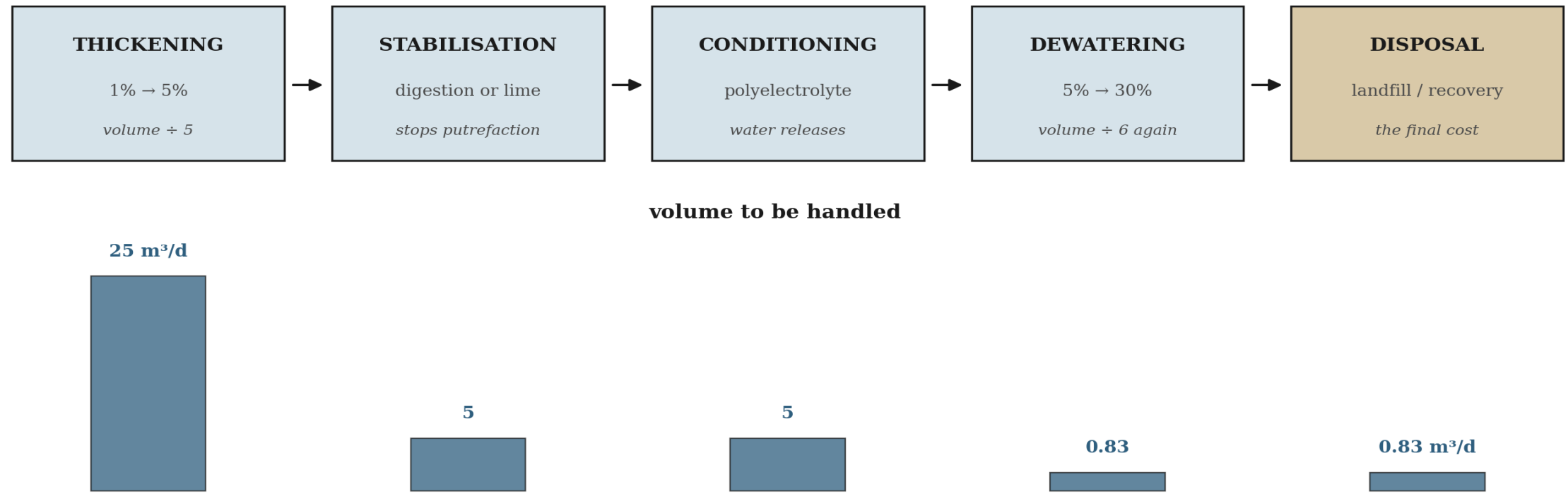
Every process in these two chapters has been described as removing pollutants from water. With the partial exception of oxidation, none of them destroys anything. They concentrate the pollution into a much smaller volume of solids.

A treatment plant does not make pollution disappear.
It moves it from a large volume of water into a small volume of sludge.

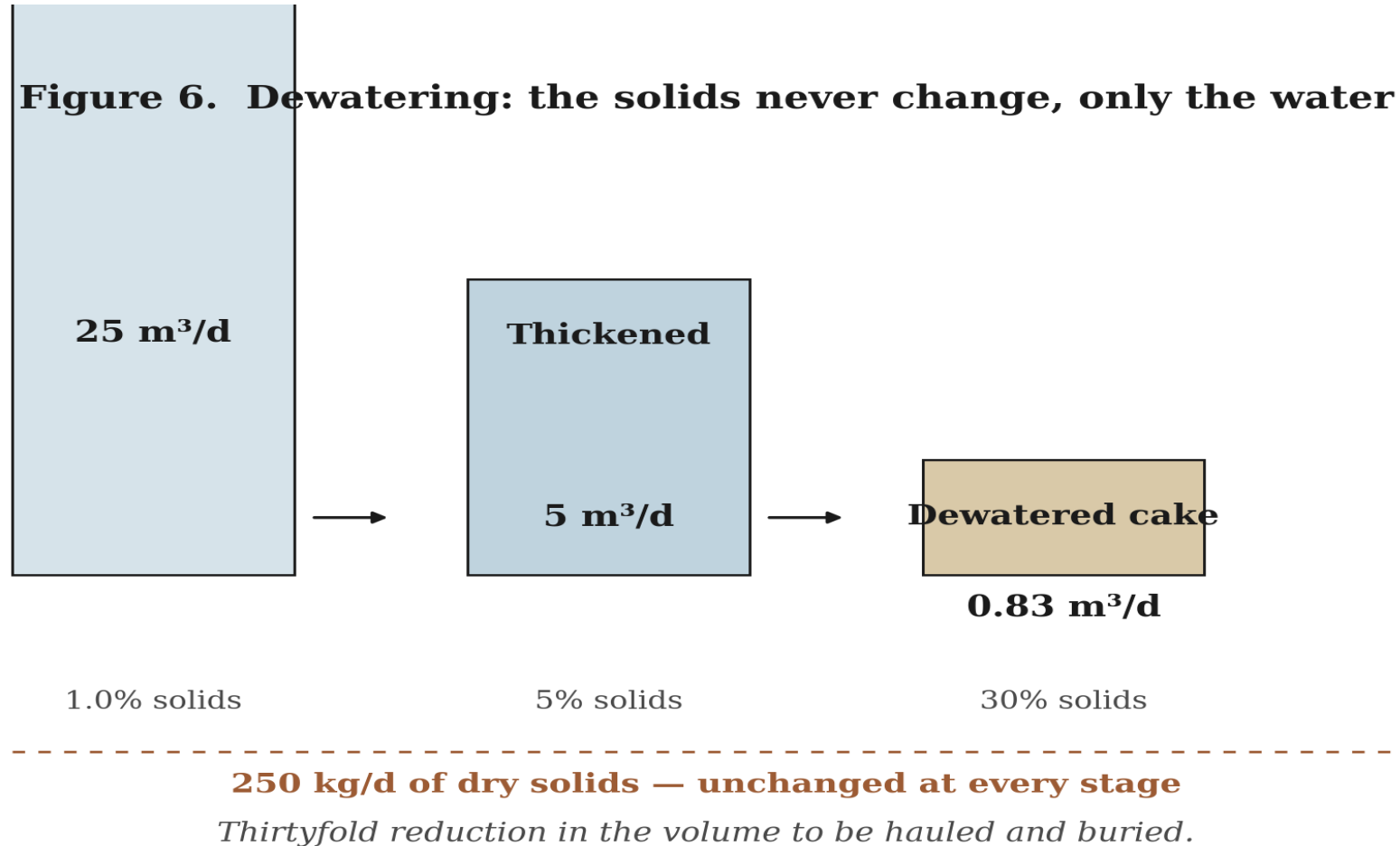
Which then becomes the plant's largest remaining problem. Sludge handling routinely accounts for a third to a half of the total operating cost of a treatment works — and in a tannery, that sludge contains the chromium.

What happens to the sludge

Figure 11. The sludge processing train



Dewatering – the arithmetic



Chrome-bearing sludge

Everything about disposal turns on one question: does the sludge contain chromium? If it does, it is hazardous waste, and the economics change entirely.

Cr(III)

the tanning form — relatively insoluble, low toxicity

Cr(VI)

soluble, mobile, toxic, carcinogenic

And careless handling converts the first into the second — during incineration at the wrong temperature, in the presence of oxidising agents, or under alkaline conditions in prolonged storage. A disposal route chosen without thought can turn a manageable waste into a dangerous one.

The same chromium is either a recovered raw material or a hazardous waste.

What decides which is a single upstream decision — whether the tanyard stream was segregated.

The other solid wastes

Fleshings, trimmings, splits and shavings are collagen. It is increasingly clear that they are raw materials that happen to be sitting in the waste yard.

Waste	Becomes
Non-chrome trimmings and splits	gelatin; collagen hydrolysate
Fleshings	fats and tallow; biogas by digestion
Hair from hair-save unhairing	keratin
Chrome shavings	protein and chromium, separated enzymatically

*Each of these turns a disposal cost into a product, and each takes load off the landfill.
This is where a leather engineer's knowledge of both the process and the waste is worth most.*

Reading

Metcalf & Eddy — Wastewater Engineering: Treatment and Resource Recovery. The main text throughout.

Crittenden et al. — MWH's Water Treatment: Principles and Design. Disinfection kinetics; membranes.

Judd — The MBR Book.

Arceivala & Asolekar — Wastewater Treatment for Pollution Control and Reuse. Subcontinental practice.

Scholz — Key Technologies for Tannery Wastewater Treatment (IULTCS); UNIDO guidance on chrome recovery and CETPs.

ECR 2023 — Bangladesh Department of Environment, discharge standards.